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Department of Artificial Intelligence and Data Science

Course Code & Name: :20AIPC503-Natural Language Processing and Chat Bot

Assignment Title: N-Grams in NLP
Assignment Number: 01
Semester: V
Sec:C

Name :
SECID :
Roll no :
Class : AI & DS 3rd year - "C"

NLP Assignment – N-Grams

Q1. Write a Python function named generate_ngrams(text, n) that:

- Takes a string of text and an integer n as input.
- Splits the text into tokens (words).
- Generates and returns a list of n-grams (as tuples).

Q2. Using your function from Q1, generate and print:

- Unigrams
- Bigrams
- Trigrams

for the text: “Natural Language Processing is a subset of AI”

Output Clear

```
Unigrams: [('Natural',), ('Language',), ('Processing',), ('is',), ('a',), ('subset',), ('of',),
    ('AI',)]
Bigrams: [('Natural', 'Language'), ('Language', 'Processing'), ('Processing', 'is'), ('is', 'a'),
    ('a', 'subset'), ('subset', 'of'), ('of', 'AI')]
Trigrams: [('Natural', 'Language', 'Processing'), ('Language', 'Processing', 'is'), ('Processing',
    'is', 'a'), ('is', 'a', 'subset'), ('a', 'subset', 'of'), ('subset', 'of', 'AI')]

=== Code Execution Successful ===
```

Q3. Choose any sentence of your choice (e.g., "The quick brown fox jumps over the lazy dog"). Using the generate_ngrams function:

1. Generate unigrams, bigrams, and trigrams.
2. Write your observations. What patterns do you notice in the output?

Output Clear

```
Unigrams: [('Tony',), ('Stark',), ('is',), ('the',), ('Marvel',), ('Jesus',)]
Bigrams: [('Tony', 'Stark'), ('Stark', 'is'), ('is', 'the'), ('the', 'Marvel'), ('Marvel', 'Jesus'
    )]
Trigrams: [('Tony', 'Stark', 'is'), ('Stark', 'is', 'the'), ('is', 'the', 'Marvel'), ('the',
    'Marvel', 'Jesus')]

=== Code Execution Successful ===
```

Q4. Extend your program to count the frequency of each n-gram in a given sentence or paragraph using Python's collections. Counter.

- Which bigram appears most frequently?
- Which trigram appears most frequently?

Output Clear

```
Most common bigram: (('Natural', 'language'), 2)
Most common trigram: (('Natural', 'language', 'processing'), 2)

=== Code Execution Successful ===
```

Q5. Answer the following reflection questions:

1. What information does an n-gram provide about a sentence?
2. How can n-grams be useful in real-world NLP applications like autocomplete or spell checking?